

EJERCICIOS INTEGRALES INDEFINIDAS

Calcular las siguientes integrales indefinidas:

1. $\int \frac{1 + \cos x}{1 - \cos x} dx$
2. $\int \frac{\operatorname{sen} x \cos x}{(1 + \cos x)^3} dx$
3. $\int \frac{1}{5 \cos^2 x + 3 \operatorname{sen}^2 x} dx$
4. $\int \frac{1}{3 - 2 \cos x} dx$
5. $\int \frac{2 - \operatorname{sen} x}{2 + \operatorname{sen} x} dx$
6. $\int \frac{1}{1 + \operatorname{tg} x} dx$
7. $\int \frac{\cos^3 x \operatorname{sen} x}{1 + \operatorname{sen}^2 x} dx$
8. $\int \operatorname{sen}(9x) \cos(4x) dx$
9. $\int \operatorname{sen}(8x + 3) \operatorname{sen}(5x - 4) dx$
10. $\int \cos(5x + 9) \cos(7x - 2) dx$

Soluciones:

1. $-x - 2 \cotg \left(\frac{x}{2} \right) + C$
2. $\frac{\frac{1}{2} + \cos x}{(1 + \cos x)^2} + C$
3. $\frac{1}{\sqrt{15}} \operatorname{arctg} \left(\frac{\sqrt{3}}{\sqrt{5}} \operatorname{tg} x \right) + C$
4. $\frac{2}{\sqrt{5}} \operatorname{arctg} \left(\sqrt{5} \operatorname{tg} \left(\frac{x}{2} \right) \right) + C$
5. $-x - \frac{8}{\sqrt{3}} \operatorname{arctg} \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \operatorname{tg} \left(\frac{x}{2} \right) \right) + C$
6. $\frac{x}{2} + \frac{1}{2} \ln |\operatorname{sen} x + \cos x| + C$
7. $-\frac{\operatorname{sen}^2 x}{2} + \ln(1 + \operatorname{sen}^2 x) + C$
8. $-\frac{1}{10} \cos(5x) - \frac{1}{26} \cos(13x) + C$
9. $-\frac{1}{26} \operatorname{sen}(13x - 1) + \frac{1}{6} \operatorname{sen}(3x + 7) + C$
10. $\frac{1}{24} \operatorname{sen}(12x + 7) + \frac{1}{4} \operatorname{sen}(2x - 11) + C$